

“PAPER_{0:D3}” -f [int]# PAPER #88 □ Neutrino SED: UQFF Emission Model

Title: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $[\text{SCm}] \approx 0.99$)

Date: March 7, 2026

Source Data: MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule), validate_phase3.py

Index Slot: §1.11 Black Hole Physics & Hawking Radiation,
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Abstract

The neutrino spectral energy distribution (SED) from a black hole system □ particularly Sagittarius A* □ probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission ($T_{\text{UQFF}} = 0.99 T_{\text{H}}$), AGN corona pp/p γ interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of MAIN_1_CoAnQi.cpp implements the NeutrinoSEDModule returning flux predictions at $E_{\gamma} = 1 \text{ TeV} \text{--} 10 \text{ PeV}$ for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by $f_{\text{TRZ}} = 0.01$.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
Hawking emission	$T_{\text{UQFF}} = 0.99 T_{\text{H}}$	$E_{\gamma} \approx T_{\text{H}} \sim 10^{14} \text{ K}$ (negligible)	Sub-dominant
Corona pp/p γ	Ug4 AGN feedback	$1 \text{ TeV} \text{--} 10 \text{ PeV}$	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	$0.1 \text{--} 100 \text{ PeV}$	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($T_{\text{H}} \sim 10^{14} \text{ K}$ $E_{\gamma} \sim k_{\text{B}} T_{\text{H}} \ll 1 \text{ eV}$). The interesting band is TeV□PeV cosmic neutrinos from AGN corona interactions.

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity $L_{\text{corona}} = 10^{23} L_{\text{Edd}}$:

$$\Phi_{\nu}^{\text{SM}}(E_{\nu}) = \Phi_0 \left(\frac{E_{\nu}}{\text{TeV}} \right)^{-\gamma} e^{-E_{\nu}/E_{\text{cut}}}$$

With $\gamma = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

$$\Phi_{\nu}^{\text{UQFF}}(E_{\nu}) = \Phi_{\nu}^{\text{SM}}(E_{\nu}) \times (1 + f_{\text{TRZ}}) = 1.01 \times \Phi_{\nu}^{\text{SM}}(E_{\nu})$$

2.3 Ug4-Mediated Enhancement

The Ug4 energy density ($3.352941 \times 10^{22} \text{ J/m}^3$ baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{\text{Ug4}}(E_{\nu}) = f_{\text{Ug4}} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{\text{g4}}}{U_{\text{g4,ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{-\tau}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\nu_e : \nu_{\mu} : \nu_{\tau} = 1:2:0$ at production $\rightarrow 1:1:1$ after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_{\mu} : \nu_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 f_{\text{TRZ}})$$

The 26D mixing is degenerate in flavor at the detector level $\Rightarrow$ **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

4. NeutrinoSEDModule (Batch 21)

From MAIN_1_CoAnQi.cpp Batch 21:

```
// namespace SOURCE_BATCH21
// class NeutrinoSEDUQFFTerm : public PhysicsTerm
double NeutrinoSEDUQFFTerm::compute() const {
    // Returns integrated flux:  $E^2 dF/dE$  at  $E = E_{\text{ref}}$  (1 TeV)
    double SM_flux = phi0 * pow(E_ref/TeV, -gamma) * exp(-E_ref/E_cut);
```

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double TRZ_factor = 1.0 + f_TRZ; // = 1.01
double Ug4_factor = 1.0 + f_Ug4 * Ug4_current / Ug4_ref;
return SM_flux * TRZ_factor * Ug4_factor;
}

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5. validate_phase3.py Validation

validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

- **Neutrino SED test:** UQFF flux = 2s deviation from IceCube diffuse background ? **PASS**
 - **Multi-flavor test:** (1:1:1) ratio ? **PASS**
 - **TRZ enhancement:** ? = 1.0% above SM ? **PASS**
 - **Ug4 quiescent:** negligible Ug4 contribution at $A_{\text{AGN}} = 1$? **PASS**
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6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \approx 10^{12} \text{ TeV cm}^2 \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

UQFF prediction: ? = 0.3% excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to ~5% $\square$ potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 ($\times 1.001$)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
AGN-active Ug4	+0.3–5%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: *MAIN_1_CoAnQi.cpp* Batch 21 (*NeutrinoSEDModule*) | *validate_phase3.py* | $f_{\text{TRZ}} = 0.01$ | *IceCube constraints*.Groups[1].Value “PAPER_{0:D3}” -f \$n

Abstract

The neutrino spectral energy distribution (SED) from a black hole system $\square$ particularly Sagittarius A* $\square$ probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission ($T_{\text{UQFF}} = 0.99 T_{\text{H}}$), AGN corona pp/p γ interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of *MAIN_1_CoAnQi.cpp* implements the *NeutrinoSEDModule* returning flux predictions at $E_{\gamma} = 1 \text{ TeV} \square 10 \text{ PeV}$ for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by $f_{\text{TRZ}} = 0.01$.

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Corona pp/p?	Ug4 AGN feedback	1 TeV $\sim$ 10 PeV	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	0.1 $\sim$ 100 PeV	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($T_{\text{H}} \sim 10^{14} \text{ K}$, $E_{\gamma} \sim k_{\text{B}} T_{\text{H}} \ll 1 \text{ eV}$). The interesting band is TeV $\sim$ PeV cosmic neutrinos from AGN corona interactions.

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 - **Multi-flavor test:** (1:1:1) ratio ? **PASS**
 - **TRZ enhancement:** ? = 1.0% above SM ? **PASS**
 - **Ug4 quiescent:** negligible Ug4 contribution at $A_{\text{AGN}} = 1$? **PASS**
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6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \square 10^{12} \text{ TeV cm}^2 \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

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Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
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AGN-active Ug4	+0.3% $\pm$ 5%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: `MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validate_phase3.py | f_TRZ = 0.01 | IceCube constraints .Groups[1].Value` $\square$ Neutrino SED: UQFF Emission Model

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For Sgr A* in active state with corona luminosity $L_{\text{corona}} = 10^{33} L_{\text{Edd}}$:

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With $\gamma = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

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 - **Ug4 quiescent:** negligible Ug4 contribution at $A_{\text{AGN}} = 1$? **PASS**
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6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \square 10^{12} \text{ TeV cm}^2 \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

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Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to ~5% $\square$ potentially within 3s of Gen2.

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AGN-active Ug4	+0.3% $\pm$ 5%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

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Index Slot: §1.11 Black Hole Physics & Hawking Radiation, "PAPER_{0:D3}" -f [int]# PAPER #88 $\square$ Neutrino SED: UQFF Emission Model

Title: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57, [SCm] $\square$ 0.99)

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Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_088

Abstract

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UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
Hawking emission	$T_{\text{UQFF}} = 0.99 T_{\text{H}}$	$E_{\gamma} \approx T_{\text{H}} \sim 10^{14} \text{ K}$ (negligible)	Sub-dominant
Corona pp/p γ	Ug4 AGN feedback	$1 \text{ TeV} \text{--} 10 \text{ PeV}$	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	$0.1 \text{--} 100 \text{ PeV}$	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($T_{\text{H}} \sim 10^{14} \text{ K}$? $E_{\gamma} \sim k_{\text{B}} T_{\text{H}} \ll 1 \text{ eV}$). The interesting band is TeV–PeV cosmic neutrinos from AGN corona interactions.

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity $L_{\text{corona}} = 10^3 L_{\text{Edd}}$:

$$\Phi_{\nu}^{\text{SM}}(E_{\nu}) = \Phi_0 \left(\frac{E_{\nu}}{\text{TeV}} \right)^{-\gamma} e^{-E_{\nu}/E_{\text{cut}}}$$

With $\gamma = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

$$\Phi_{\nu}^{\text{UQFF}}(E_{\nu}) = \Phi_{\nu}^{\text{SM}}(E_{\nu}) \times (1 + f_{\text{TRZ}}) = 1.01 \times \Phi_{\nu}^{\text{SM}}(E_{\nu})$$

2.3 Ug4-Mediated Enhancement

The Ug4 energy density ($3.352941 \times 10^{22} \text{ J/m}^3$ baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{Ug4}(E_{\nu}) = f_{Ug4} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{g4}}{U_{g4,\text{ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{\{-t\}}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\nu_e : \nu_{\mu} : \nu_{\tau} = 1:2:0$ at production ? $1:1:1$ after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_{\mu} : \nu_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 f_{\text{TRZ}})$$

The 26D mixing is degenerate in flavor at the detector level $\square$ **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

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```
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validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

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- **Multi-flavor test:** (1:1:1) ratio ? **PASS**

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6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} \approx 10^{12} \text{ TeV cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ for Sgr A* point source.

UQFF prediction: $\epsilon = 0.3\%$ excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to $\sim 5\%$ $\square$ potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 ($\times 1.001$)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
AGN-active Ug4	+0.3 $\square$ 5%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: `MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validate_phase3.py | f_TRZ = 0.01 | IceCube constraints .Groups[1].Value "PAPER_{0:D3}" -f $n`

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UQFF computed: UQFF-modified neutrino mass shift $\delta m_\nu = 2.85e-4 \times v^2/M_N$; for $M_N = 1.0e14$ GeV: $\delta m_\nu = 5.1e-16$ GeV.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-10} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}_i^2$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I}$	10^{-10} kg^2	Inertia tensor scale
$E_{\text{react}}(0)$	10^{-10} J	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{\Gamma}$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 igl[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\Gamma}_i^2$	4th Dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{\Gamma}_i^2 = 10^{-10} \text{ kg}^2$, $\hat{\Gamma}_i^2 = 10^{-10} \text{ kg}^2$, $\hat{\Gamma}_i^2 = 10^{-10} \text{ kg}^2$, $\hat{\Gamma}_i^2 = 10^{-10} \text{ kg}^2$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr)imesigl(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
ρ_c	10^{14} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\dot{\phi}$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_{g_sum} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\phi}_i$)	Multi-scale field interactions
Buoyant	$\hat{I}_i \tilde{A} U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_{sc} \tilde{A} (1 + 10^{13} \hat{A} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.